Two Negative Results for Vector Symbolic Architectures: FFN Replacement and Compositional Image Generation

Peter C.
Independent Researcher
peterc3@live.com

Abstract

Vector Symbolic Architectures (VSAs) offer algebraically structured representations—binding, bundling, permutation—that are $O(D)$ and matmul-free. These properties make VSA a natural candidate for (1) replacing dense feed-forward networks with structured retrieval, and (2) compositional image generation via token binding. We conduct systematic experiments testing both hypotheses and find that neither works, for different but complementary reasons.

Case Study 1: FFN Replacement. We test VSA memory layers as replacements for feed-forward networks in Qwen3.6-27B. The failure mode is a rank bottleneck: VSA's cleanup→bind→retrieve pipeline has effective rank bounded by top-k (typically 4), while FFN mappings are ∼89% linear with effective rank >2048. A 164K-parameter rank-16 linear projection captures more variance than a 35M-parameter VSA memory layer.

Case Study 2: Compositional Image Generation. We test VSA binding for encoding image token sequences from TiTok VQ-8K (64 tokens × 8192 codebook). The failure has two independent causes: (a) FHRR superposition of 64 bindings cannot support 8192-way retrieval (0% accuracy at all D tested), and (b) real images do not share token multisets (<9% overlap), so the permutation-based generation framing is inapplicable.

Positive findings. We identify conditions where VSA succeeds: permutation locality in TiTok, 100% exact permutation recovery at $D{\geq}2048$, and SVD-aligned codebooks capturing 50% more variance than random. These successes illuminate the boundary: VSA excels at structured, low-cardinality operations but fails when the target requires high-rank continuous mappings or high-cardinality discrete retrieval.

1 Introduction

Vector Symbolic Architectures (VSAs), also known as hyperdimensional computing, have attracted renewed interest as a biologically plausible and algebraically structured alternative to standard neural network operations [Kanerva, 2009, Plate, 2003]. VSA represents data as high-dimensional vectors and computes with three core operations: *binding* (element-wise multiplication), *bundling* (superposition via addition), and *permutation* (cyclic shift). These operations are $O(D)$, matmul-free, and support exact inversion—properties that are absent in standard dense layers.

Two applications stand out as natural fits for VSA's strengths:

1. **FFN replacement.** Feed-forward networks consume two-thirds of transformer parameters and are known to store factual knowledge as key-value associations [Sukhbaatar et al., 2019]. VSA's bind-and-retrieve mechanism is a direct analog of associative memory, suggesting that VSA memory layers could replace dense FFNs with structured retrieval at lower cost.

2. **Compositional image generation.** Image tokenizers like TiTok [Yu et al., 2024] encode images as sequences of discrete tokens. VSA binding naturally represents position→content mappings, and permutation operations could support compositional manipulation of token arrangements.

We test both hypotheses with systematic experiments and find that *neither works*, for different but complementary reasons.

FFN replacement fails due to a **rank bottleneck**: VSA's top-k retrieval constrains output rank to $\leq k$, while FFN effective rank exceeds 2048.

Image generation fails due to a **capacity bottleneck** (64 superposed bindings cannot support 8192-way retrieval) *and* a **framing mismatch** (images do not share token multisets, invalidating the permutation premise).

Positive findings narrow the scope: VSA achieves 100% exact match for permutation encoding at $D \geq 2048$ (64-way classification), confirming that VSA works for structured, low-cardinality operations.

Contributions.

1. Two independent case studies demonstrating VSA limitations in complementary domains.
2. A rank-bottleneck diagnosis for FFN replacement (theoretical proof sketch + empirical confirmation).
3. Capacity analysis for VSA token binding: theory predicts $\sim \log_2(D/N)$ bits per retrieval; experiments confirm.
4. Positive result: VSA permutation encoding achieves 100% exact match at D=2048 for 64-position permutations.
5. Negative result: TiTok VQ-8K token multisets have <9% overlap between images, invalidating permutation-based generation.

2 Background

2.1 Vector Symbolic Architectures

We use FHRR (Frequency Holographic Reduced Representation) [Plate, 2003], where each symbol is a D-dimensional complex phasor $\mathbf{x} = e^{i\boldsymbol{\theta}}$ with $\boldsymbol{\theta} \in [-\pi, \pi)^D$. Binding is element-wise complex multiplication: $\text{bind}(\mathbf{a}, \mathbf{b}) = \mathbf{a} \odot \mathbf{b}$. Bundling is complex addition followed by normalization: $\text{bundle}(\mathbf{a}, \mathbf{b}) = \text{normalize}(\mathbf{a} + \mathbf{b})$. Unbinding uses the conjugate: $\text{unbind}(\mathbf{c}, \mathbf{a}) = \mathbf{c} \odot \mathbf{a}^*$.

The storage capacity for N superposed bindings is approximately $\log_2(D/N)$ bits per retrieval [Frady et al., 2018]. This means that with N=64 bindings and D=4096, each retrieval can distinguish among at most $2^6 = 64$ items—far below the 8192-entry codebook (2^{13}) needed for image token retrieval.

2.2 Feed-Forward Networks in Transformers

Standard transformer FFNs consist of an up-projection, nonlinear activation, and down-projection: $\text{FFN}(\mathbf{x}) = W_2 \cdot \sigma(W_1\mathbf{x})$. These layers contain $2 \times d_{\text{model}} \times d_{\text{ffn}}$ parameters per layer and are known to store factual knowledge [Sukhbaatar et al., 2019].

2.3 Image Tokenization with TiTok

TiTok [Yu et al., 2024] is a 1D image tokenizer that encodes a 256×256 image into 64 discrete tokens from an 8192-entry codebook (64 dimensions per entry). We use the frozen pre-trained `titok_bl64_vq8k_imagenet` model (Apache 2.0 license) throughout.

3 Case Study 1: VSA for FFN Replacement

3.1 Setup

We extract activation pairs $(x_{\text{in}}, x_{\text{out}})$ from Qwen3.6-27B [Qwen Team, 2024] (d_{model}=5120, 64 layers) at layers 3, 27, and 43. We sample 50,000 tokens from FineWeb-Edu with 5% held out for validation. The metric is *variance captured*: $\text{Var\%} = (1 - \text{MSE}_{\text{val}}/\text{MSE}_{\text{zero}}) \times 100\%$, where MSE_{zero} is the mean squared error of predicting zero.

3.2 Architectures Tested

- **VSA V1** (frozen random codebooks): standard FHRR bind-retrieve with random position and content phasors.
- **VSA V2** (learned + Gumbel-softmax): codebook entries and routing learned end-to-end.
- **VSA-MoE**: VSA routing selects experts in a mixture-of-experts architecture.
- **Baselines**: rank-r linear (SVD), rank-r + MLP, learned-gate MoE.

3.3 Results

Table 1 shows the architecture comparison at layer 27. The key finding: a 164K-parameter rank-16 linear projection (5.9%) *outperforms* a 35M-parameter VSA memory layer (4.8%). Rank-2048 linear (36.6%) dominates all VSA variants by 7× or more.

Table 1: Architecture comparison (layer 27, Qwen3.6-27B).

Architecture	Var%	Params
Zero baseline	0.0	0
Cube Memory V1 (frozen VSA)	∼5	35M
Cube Memory V2 (learned VSA)	4.8	35M
Rank-16 linear (SVD)	5.9	164K
VSA-MoE 16×128 top-4	14.2	24M
Learned-MoE 8×256 top-2	16.2	21M
Rank-2048 linear	36.6	21M
Rank-2048 + MLP-512	**38.4**	**26M**
Full-rank ceiling	41.1	52M

3.4 Ablation Studies

Top-k scaling. Increasing top-k from 4 to 64 does not break the ceiling: variance captured decreases slightly from 2.9% to 2.7%. This rules out the hypothesis that the bottleneck is insufficient retrieval breadth.

Codebook ablation. SVD-aligned codebooks improve over random (4.3% vs 2.8%), confirming that aligning codebook entries with the data manifold helps. However, the improvement does not change the conclusion: all VSA variants remain far below linear baselines.

FLOPs comparison. All FLOPs are inference-only forward pass. Rank-2048 linear achieves 36.6% variance at 42M FLOPs/token. VSA V2 achieves 4.8% at 62M FLOPs/token—*worse and slower*.

Rank-equalized comparison. At matched effective rank 4, a rank-4 linear (SVD) yields negative R^2 (worse than predicting the mean) at 41K FLOPs, while VSA top-4 achieves 2.9% at 62M FLOPs. VSA's marginal advantage comes from data-dependent routing (selecting different codebook entries per input), but both are catastrophically below rank-2048 linear (36.6%). VSA spends 1514× more FLOPs for this marginal advantage.

3.5 Diagnosis: The Rank Bottleneck

The VSA retrieval output is:

$$\mathbf{y} = \sum_{j \in \text{top-}k} \alpha_j \cdot \mathbf{v}_j, \quad \alpha_j = \text{softmax}(\text{sim}_j), \tag{1}$$

where $\mathbf{v}_j$ are memory value vectors. This is a convex combination of k vectors, so $\mathbf{y} \in \text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$, giving rank(output) $\leq k$.

With top-k=4, VSA output rank is at most 4. The FFN effective rank exceeds 2048 (singular values decay slowly; see Appendix). Retrieval is fundamentally underpowered: the FFN mapping is $\sim$89% linear, and the remaining $\sim$11% is better captured by a small MLP than by sparse retrieval.

This rank bottleneck applies to *all* retrieval-based FFN replacements—Product Key Memory [Lample et al., 2019], holographic reduced representations, or any top-k selection mechanism.

4 Case Study 2: VSA for Image Generation

4.1 Motivation: Permutation-Group Indexed Generation

We hypothesize that image generation can be framed as token permutation manipulation. TiTok encodes images as 64 discrete tokens from an 8192-entry codebook. VSA binding naturally represents position→content mappings: $\mathbf{h} = \sum_{i=1}^{64} \text{bind}(\mathbf{p}_i, \mathbf{c}_{t_i})$, where $\mathbf{p}_i$ is a position phasor and $\mathbf{c}_{t_i}$ is a content phasor for token t_i.

4.2 Prerequisite: Permutation Locality (Experiment 0)

We first validate that the TiTok decoder is smooth with respect to token permutation. We encode a test image, apply N random position swaps, decode, and measure MSE against the original.

The monotonic relationship (Table 2) confirms the generation premise has empirical support: small permutations yield small visual changes.

Table 2: Permutation locality: MSE vs. number of random token swaps.

Swaps	0	1	4	16	64
MSE	0.000	0.001	0.022	0.073	0.153

4.3 Token Binding Fails (Experiments 1, 1b, 1c, 2)

Experiment 1: Pure VSA binding. We bind 64 position phasors with content phasors, superpose, unbind each position, and classify against the 8192-entry codebook. Theory predicts $\sim\log_2(D/64)$ bits per retrieval; 8192 entries need 13 bits. Even D=4096 provides only 6 bits. Result: 0% accuracy at all D tested (Table 3).

Table 3: Pure VSA token binding: accuracy vs. dimension.

D	Acc%	Bits (have / need)
512	0.0	3.0 / 13.0
1024	0.0	4.0 / 13.0
2048	0.0	5.0 / 13.0
4096	0.0	6.0 / 13.0

Experiment 1b: VSA + MLP decoder. Adding a 2-layer MLP after unbinding produces catastrophic overfitting at all D: training loss approaches zero while validation loss reaches 25. The decoder memorizes training-specific noise patterns. An MLP-only control (position one-hot → MLP → 8192-way, no VSA) achieves 0.93% accuracy, confirming the task is impossible from positional information alone.

Experiment 1c: Factored codebook. We decompose $8192 = 128 \times 64$ and use 3-way binding (pos $\otimes$ factA $\otimes$ factB), retrieving factors independently. A hard ceiling appears at $\sim$5.8% for the 64-way factor regardless of D (Table 4). The bottleneck is structural (cross-talk from superposition), not capacity.

Table 4: Factored codebook (128×64): accuracy per factor.

D	AccA% (128-way)	AccB% (64-way)	Joint%
512	2.2	5.6	0.1
1024	2.3	5.8	0.2
2048	2.0	5.7	0.1
4096	1.9	5.9	0.1

Experiment 2: Real TiTok tokens. We test on 6000 Imagenette images encoded by frozen TiTok. Token distribution entropy is 12.85 / 13.00 bits; all 8192 codes are used. At D=4096, accuracy plateaus at 0.61% after 1500 steps—the same plateau as D=512. Near-uniform codebook usage means real tokens are as hard as random.

4.4 Permutation Encoding Works (Experiment 3)

We reframe: encode *permutations* (64-way classification) instead of *tokens* (8192-way). This is a pure VSA theory test requiring no training. For each permutation σ, we encode $\mathbf{h}_\sigma = \sum_{i=1}^{64} \mathrm{bind}(\mathbf{p}_i, \mathbf{p}_{\sigma(i)})$ and decode by unbinding each source position and classifying among 64 destinations.

Table 5: Permutation recovery: exact match % (n=2000 per condition).

D	k=1	k=2	k=4	k=8	k=16	k=32
256	0.0	0.0	0.0	0.0	0.0	0.3
512	0.0	0.0	0.0	0.0	0.0	67.2
1024	89.3	76.3	54.6	29.6	15.3	99.9
2048	99.8	99.8	99.8	98.6	96.5	**100.0**
4096	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

At $D \geq 2048$, recovery is perfect (0 failures in 2000 test permutations per condition, 95% CI: [99.8%, 100%]). The 64-way classification is well within VSA capacity, confirming that VSA excels at structured operations on small discrete sets.

4.5 But Images Aren't Permutations (Experiment 4)

We test whether real images can be approximated as permutations of a cluster reference. For each image: find nearest cluster (by token histogram similarity, k-means), solve optimal assignment via the Hungarian algorithm [Kuhn, 1955], and measure token match rate.

Table 6: Permutation reconstruction: can images be expressed as permutations of references?

K clusters	Within-cluster overlap	Match rate	$\geq$50% match
10	4.0%	2.7%	0.2%
50	2.1%	5.2%	0.8%
200	2.4%	8.9%	3.3%

"Within-cluster overlap" is the average pairwise multiset intersection size divided by 64 between random pairs in the same cluster. "Match rate" is the fraction of positions where the Hungarian-optimal permutation of the reference yields the correct token.

Each image uses $64/8192 = 0.78\%$ of the codebook. Under uniform usage, the expected multiset overlap between two random images is $\sim$0.5 tokens (birthday collision rate: $8192 \times (64/8192)^2 \approx 0.5$). The permutation framing requires near-complete multiset overlap, which is impossible when the codebook is large and near-uniformly used.

4.6 Diagnosis: Two Independent Failure Modes

1. **Capacity failure.** FHRR superposition of N bindings provides $\sim \log_2(D/N)$ bits per retrieval. With N=64 and codebook size 8192 (13 bits), no feasible D suffices. Reducing cardinality to 64-way (permutations) solves this.

2. **Framing failure.** Even if token retrieval worked, permutation-based generation requires shared token multisets between images. With 8192 near-uniformly-used codebook entries, this assumption fails catastrophically (<9% overlap).

5 Positive Findings and Boundary Conditions

5.1 Where VSA Succeeds

1. **Permutation representation.** 100% exact recovery at D=2048 for 64-position permutations (Experiment 3). VSA is ideal for structured operations on small discrete sets.

2. **Permutation locality.** The TiTok decoder is smooth with respect to token permutation (Experiment 0). This property is real and potentially useful for local image editing.

3. **SVD-aligned codebooks.** Initializing VSA codebooks from SVD of the target activation matrix captures 50% more variance than random (4.3% vs 2.8%).

5.2 The Boundary

VSA succeeds when classification cardinality is low (64 positions, not 8192 codes), the target has discrete compositional structure, and operations are group-theoretic. VSA fails when the target mapping is high-rank and continuous, retrieval requires high-cardinality discrimination from superposition, or the compositional framing does not match the data.

5.3 Generalization

The rank bottleneck (Case Study 1) applies to *all* retrieval-based FFN replacements: Product Key Memory [Lample et al., 2019], Memory Layers [Wu et al., 2024], or any top-k selection mechanism. The output of any top-k retrieval is constrained to a k-dimensional subspace, which is insufficient for the high-rank mappings that FFNs implement.

The capacity limit (Case Study 2) applies to any VSA scheme that superposes N bindings and retrieves from codebook size C: reliable retrieval requires $D \gg N \times C$.

6 Related Work

VSA and Hyperdimensional Computing. The theoretical foundations of VSA trace to Kanerva's sparse distributed memory [Kanerva, 2009] and Plate's holographic reduced representations [Plate, 2003]. Frady et al. [2018] established capacity bounds for FHRR, which our experiments confirm empirically.

Memory Layers. Product Key Memory [Lample et al., 2019] and recent scaling work [Wu et al., 2024, Sukhbaatar et al., 2019] use key-value retrieval to augment or replace FFN layers. Our rank bottleneck analysis applies to these methods whenever they use top-k selection.

FFN Compression. Low-rank factorization [Hsu et al., 2022], pruning [Frantar and Alistarh, 2023], and distillation [Hinton et al., 2015] are alternatives to memory-based replacement. Our results show that even simple rank-2048 linear projections dramatically outperform structured retrieval approaches.

Image Tokenization. VQ-VAE [van den Oord et al., 2017], VQGAN [Esser et al., 2021], and TiTok [Yu et al., 2024] have established discrete tokenization as a viable representation for image generation. Our permutation locality finding (Experiment 0) adds a new characterization of TiTok's latent space.

Sparse MoE. Mixture-of-experts approaches [Shazeer et al., 2017, Fedus et al., 2022, Jiang et al., 2024] use routing to select among specialized sub-networks. Our VSA-MoE variant (14.2%) underperforms learned-gate MoE (16.2%), suggesting that VSA routing provides no advantage over standard gating.

7 Conclusion

We tested VSA in two settings where its algebraic properties should provide advantages: structured FFN replacement and compositional image generation. Both fail, for different reasons:

1. **FFN replacement: the rank bottleneck.** FFN mappings are ∼89% linear with effective rank >2048. VSA retrieval collapses rank to top-k. Linear projections dominate at every parameter budget.

2. **Image generation: capacity bottleneck plus framing mismatch.** Superposition of 64 bindings provides ∼6 bits per retrieval; 8192 codebook entries need 13 bits. And even with perfect retrieval, images do not share token multisets, so the permutation framing fails.

The positive findings narrow VSA's useful scope: it excels at representing discrete group operations (permutations of 64 positions: 100% exact match) but fails when the target requires either high-rank continuous mappings or high-cardinality discrete retrieval.

These results suggest that VSA's future in deep learning lies in genuinely compositional, low-cardinality tasks—symbolic reasoning, relational learning, discrete program synthesis—rather than as a drop-in replacement for dense continuous computations.

All code and data are available at `https://github.com/Peterc3-dev/cube-memory`.

References

Patrick Esser, Robin Rombach, and Bjorn Ommer. Taming transformers for high-resolution image synthesis. In *CVPR*, 2021.

William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *JMLR*, 2022.

E. Paxon Frady, Denis Kleyko, and Friedrich T. Sommer. A theory of sequence indexing and working memory in recurrent neural networks. In *Neural Computation*, volume 30, pages 1449–1513, 2018.

Elias Frantar and Dan Alistarh. SparseGPT: Massive language models can be accurately pruned in one-shot. In *ICML*, 2023.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Yen-Chang Hsu, Ting Hua, Sungen Chang, Qian Lou, Yilin Shen, and Hongxia Jin. Language model compression with weighted low-rank factorization. *arXiv preprint arXiv:2207.00112*, 2022.

Albert Q. Jiang, Alexandre Sablayrolles, Antoine Rouzaud, et al. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.

Pentti Kanerva. Hyperdimensional computing: An introduction to computing in distributed representation with high-dimensional random vectors. *Cognitive Computation*, 1(2):139–159, 2009.

Harold W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1-2):83–97, 1955.

Guillaume Lample, Alexandre Sablayrolles, Marc'Aurelio Ranzato, Matti Ott, and Armand Joulin. Large memory layers with product keys. In *NeurIPS*, 2019.

Tony A. Plate. Holographic reduced representations. *IEEE Transactions on Neural Networks*, 2003.

Qwen Team. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2024.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *ICLR*, 2017.

Sainbayar Sukhbaatar, Edouard Grave, Guillaume Lample, Herve Jegou, and Armand Joulin. Augmenting self-attention with persistent memory. In *NeurIPS*, 2019.

Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *NeurIPS*, 2017.

Vincent-Pierre Wu, Sainbayar Sukhbaatar, and Guillaume Lample. Memory layers at scale. *arXiv preprint arXiv:2412.09764*, 2024.

Qihang Yu, Jose Lezama, Daniel Cremers, Jia Liang, Hao Lu, Oncel Tuzel, Aysegul Karagoz, and Jiahui Jia. An image is worth 32 tokens for reconstruction and generation. In *NeurIPS*, 2024.

A Experimental Details

Hardware. All Case Study 2 experiments ran on an AMD Ryzen AI 9 HX 370 APU with 23 GB RAM (no discrete GPU). Case Study 1 activation extraction used a ThinkCentre M70q Gen 5 with 32 GB RAM. Total compute: approximately 8 hours across both machines.

Hyperparameters. VSA experiments used AdamW with learning rate 10^{-3}, weight decay 0.01, cosine annealing, batch size 16 (64 for FFN distillation), and 2000–3000 training steps. Position phasors were frozen random; content phasors were learned.

Reproducibility. All experiments use fixed random seeds (42). The TiTok model is the publicly available `yucornetto/tokenizer_titok_bl64_vq8k_imagenet` checkpoint. Imagenette 320 was used for image tokenization (6000 images: 5000 train, 1000 val).